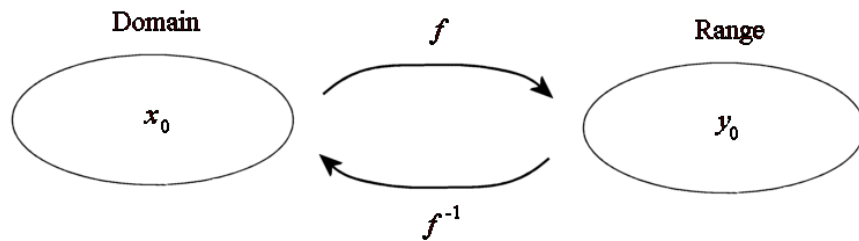


Section 1.4

I. Inverse functions

Function f : {Domain= D , Rule, Range= R }

Inverse f^{-1} : {Domain= R , Reverse Rule, Range= D }



Definition

Function	Inverse
$y = f(x)$	$y = f^{-1}(x) \Leftrightarrow x = f(y)$

II. Solving for the inverse function

Examples

III. Criteria for the existence of the inverse function: Horizontal Line Test

Theorem 1.2 *A strictly monotonic function has an inverse*

Procedure for graphing the inverse function: Reflection about the line $y = x$

Examples

IV. Trigonometric Functions:

Require domain restrictions to have inverses

Notation

A. Sine $y = \sin x$ $y = \sin^{-1} x$

Function: Domain _____ Range _____

Inverse: Domain _____ Range _____

Graphs

B. Cosine $y = \cos x$ $y = \cos^{-1} x$

Function: Domain _____ Range _____

Inverse: Domain _____ Range _____

Graphs

C. Tangent $y = \tan x$ $y = \tan^{-1} x$

Function: Domain _____ Range _____

Inverse: Domain _____ Range _____

Graphs

E. Properties

1. Evaluation
2. Inversion
3. Identities
4. Relationships

Examples